PIF-Net: A Deep Point-Image Fusion Network for Multimodality Semantic Segmentation of Very High-Resolution Imagery and Aerial Point Cloud

Zhou Guo, Rui Xu, Chen-Chieh Feng, and Zhao Zeng

***Abstract*— Semantic segmentation is of great significance in many applications. However, automating such a task on single-modality data is challenging in the field of remote sensing due to complex scenes, occlusions, and homogeneous data. In this article, we propose a deep point-image fusion network, namely PIF-Net, for multimodality semantic segmentation. The proposed PIF-Net encompass an encoder–decoder structure, where the encoder uses two independent branches with Res-Pooling blocks and point attention (Pt-Atten) blocks to extract deep and condensed multimodal features, and the decoder upsamples these features. A hierarchical fusion module is proposed to adaptively fuse multimodal features at different levels to ensure they are fully mixed. It outputs joint features in both the point and pixel representations, which are further input to a classification module to fulfill the multiple classification tasks and obtain image and point cloud semantic segmentation results. The proposed network was tested on two benchmark datasets: the Urban Semantic 3-D (US3D) dataset and the ISPRS Vaihingen dataset. Evaluation results showed that PIF-Net achieved an overall accuracy (OA) of 91.5% and 97.2% for image and point segmentation on the US3D dataset, and an OA of 90.1% and 89.3% for image and point segmentation on the ISPRS Vaihingen dataset. Comparisons with existing single-modality and multimodality methods have indicated that PIF-Net outperformed most classic methods and could bring significant improvements. It has also demonstrated that deep multimodality learning exhibit great potentials in remote sensing applications.**

***Index Terms*— Deep learning, multimodality, point cloud, semantic segmentation, very high-resolution (VHR) imagery.**

I. Introduction

SEMANTIC segmentation is a technique of categorizing information which assigns a semantic label to each pixel or point. It is of great significance in fields of land use/land cover (LULC) extraction [1], urban environment monitoring, change detection [2], target identification [3], and 3-D modeling [4]. Semantic segmentation is usually conducted on a single type of data [5], such as very high-resolution (VHR) images [6], airborne laser scanner (ALS) point cloud data [7], and synthetic aperture radar (SAR) images [8], and have been widely researched. However, previous studies have demonstrated that semantic segmentation on a single dataset is a challenging task due to complexities of scenes, occlusions and homogeneity of the adopted data [9], [10].

In recent years, multimodality data fusion has attracted increasing attention because of the rapid development of remote sensing and geographic information system (GIS) technologies [11]. Multimodality data are captured by different sensors and have their own advantages and disadvantages [12]. Specifically, VHR images are abundant with spectral and textual information, while ALS point clouds provide 3-D information. The complementary information embedded by multimodality data makes it more efficient and robust in semantic segmentation and can achieve more accurate and better results than single-modal data [13]. Therefore, semantic segmentation with VHR imagery and ALS point clouds is becoming a hot research topic and exhibits great potentials. However, it is not easy to analyze the multimodality data jointly due to their different representations, as VHR images are represented by regular pixels, while ALS data are represented by irregular points. The different representations and spatial resolutions of multimodality data add additional challenges in data alignment and data analysis [14].

Traditional studies in fusing VHR remote sensing images and ALS point clouds can be divided into model-driven or data-driven approaches. Model-driven approaches normally make assumptions regarding characteristics of objects, such as shapes of buildings [15], geometries of roads, and spectra of vegetations. However, model-driven approaches rely heavily on user experiences and can hardly be applied to all objects of the same category due to complexities of urban environments. Data-driven approaches use the opposite strategy of their model-driven counterparts by not making any assumptions on the objects. Instead, data-driven approaches implement semantic segmentation by automatically learning features from the data themselves. Machine learning approaches are widely employed in traditional data-driven approaches, such as support vector machine (SVM) and random forest (RF). However, these approaches do not address the question of how to select appropriate features during the classification process.

With the rise of deep learning, especially the significant performance of deep learning in the field of remote sensing, deep learning-based approaches have been gradually adopted

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in multimodality data fusion. Existing studies have been found in combining SAR and optical images with deep learning models. For example, Zheng et al. [16] proposed a deep multisensory learning model using SAR images and optical images for all-weather mapping. Hong et al. [11] developed a multimodal deep learning framework for remote sensing imagery classification, in which multispectral images and SAR images were fused. Although SAR and optical images belong to multimodality data, they have very similar representations, namely, both are represented by pixels. In terms of fusing VHR optical images and ALS point clouds, their different representations (i.e., pixels and points) and dimensions (i.e., 2-D and 3-D) make it more challenging for multimodality deep learning than combining SAR and optical images.

The great potentials of deep learning-based approaches have attracted attentions in fusing VHR optical images and ALS point clouds for semantic segmentations. Poliyapram et al. [17] proposed a pointwise LiDAR and image multimodal fusion network (PMNet) for 3-D semantic segmentation of ALS point clouds. It fused aerial image features and at the same time respected basic characteristics of point clouds. However, the proposed method fused image and point cloud features by simply feature mapping and feature concatenating, while feature interaction between the two modalities was missing, resulting in ordinary segmentation results. Besides, PMNet only supported point segmentation, while segmentation results of images were not available. Xu et al. [18] proposed a range-point-voxel fusion network (RPVNet) for point cloud segmentation. It first transformed the original point cloud into a range image and voxels. Features were then extracted and fused with mutual information interactions among these views. Although the proposed network fused point clouds and range images, only one single-modality data source was used, and the proposed method cannot be generalized to normal optical images and point cloud data.

Despite the increasing interest in developing fusion networks for multimodality semantic segmentations of the VHR imagery and ALS point clouds, there are at least two challenges. First, VHR imagery and ALS point clouds are inconsistent in data representation. Most current studies adopted strategies of data transformation to unify data formats so that the multimodality deep learning with the two data sources were fulfilled. However, information loss is inevitable in either transforming point clouds to images or mapping images onto point clouds, which further result in unsatisfactory segmentation results. Second, there studies attempting to extract deep features from original images and point clouds individually using image and point cloud neural networks. However, it remains a question of how to combine the two neural networks due to their large variety in network structures and high modality discrepancies.

To tackle the challenges above, this article proposes a deep point-image fusion network (PIF-Net), aiming at improving image and point cloud segmentation results. The key idea of PIF-Net is to use a two-branch network to extract deep features of different levels from VHR imagery and ALS point clouds, and hierarchically fuse the multimodality features based on concurrent inputs. Moreover, the proposed PIF-Net interaction mechanism is highly efficient. By leveraging this hierarchical fuse strategy and efficient interaction mechanism, the proposed PIF-Net is expected to output satisfactory segmentation results of images and point clouds simultaneously. The main contributions are as follows.

1) It adopts a deep fusion framework which combines two deep learning-based branches with complementary different structures designed for images and point cloud data. The proposed method does not require data transformation before modality feature fusion, and therefore avoiding information loss to the maximum.
2) A hierarchical fusion module with an efficient interaction mechanism is designed to adaptively fuse deep features from imagery and point clouds. It ensures the multimodality features are fully mixed from low to high levels. The proposed hierarchical fusion mechanism can be generalized to other data sources.
3) Unlike the current multimodal methods that output either pixel or point labels, the proposed PIF-Net is a multitask network, which outputs semantic segmentation results of VHR imagery and ALS point clouds simultaneously.

The remainder of this article is organized as follows. Related works are introduced in Section II, while the proposed PIF-Net is described in Section III. Section IV presents the experiments and results of the multitask semantic segmentation, followed by a discussion in Section V. The conclusions are drawn in Section VI.

II. Related Works

Current multimodal deep learning studies with VHR images and LiDAR point clouds can be categorized into three approaches according to the phase of data fusion: early fusion, late fusion, and middle fusion methods.

A. Early Fusion Methods

Early fusion methods usually fuse the multimodal data at the input layer of deep learning models, and there are two strategies for doing so, i.e., mapping images onto 3-D point clouds or converting 3-D point clouds to 2-D images.

In terms of the first strategy, VHR images are mapped onto LiDAR point clouds to form colored points, and inputs the colored point clouds to deep learning models such as PointNet [19], PointNet++ [20], and PointWeb [21]. For example, Shi et al. [22] first coregistered point clouds and VHR images, and then transferred spectral features of image pixels to corresponding points. Finally, a multiscale PointNet model was employed to semantically segment multispectral point clouds.

For the second strategy, 3-D point clouds are converted to digital surface models (DSMs), which are further concatenated with multispectral images of the same spatial resolution, and form 2-D images of multiple bands. The concatenated image is input to an image segmentation network such as U-Net [23] and DeepLabv3+ [24]. For example, Kampffmeyer et al. [25] first converted LiDAR point clouds to a DSM, and concatenated the DSM, a normalized DSM and an RGB image into a five-channel image. A fully connected layer network

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was then employed to semantically segment the fused images. Huang et al. [26] concatenated multispectral images and LiDAR-derived nDSM as the input data, which were further input to a gated residual refinement network for building extraction.

Early fusion methods successfully solve multimodality segmentation problems with single-modality networks. Although they guarantee a high degree of data fusion as the models can learn deep features from input layers at the very front of the networks, information loss is unavoidable during data transformation processes. Besides, early fusion methods normally ignore modality discrepancies between point clouds and VHR images.

B. Late Fusion Methods

Late fusion methods are primarily based on two independent deep learning-based models to obtain semantic segmentation results from VHR images and LiDAR point clouds, individually. The fusion process is conducted at a late phase when both segmentation results are analyzed comprehensively to obtain a final segmentation result. For example, Bigdeli et al. [27] generated two convolution neural network-SVM (CNN-SVM) ensemble systems, one for classifying LiDAR and VHR images, and the other for classifying hyperspectral images. A decision fusion method was employed to combine the classification results from CNN ensembles, and results indicated that the proposed ensemble CNN improved single deep CNN. Hang et al. [28] fused hyperspectral and LiDAR data using two coupled CNNs. One CNN was designed to learn spectral–spatial features from hyperspectral data, while the other one was designed to capture the elevation information from LiDAR data. For the decision-level fusion, a weighted summation strategy was used to fuse the classification results of hyperspectral and LiDAR data, where the weights were determined by the classification accuracy of each output.

Although late fusion methods are easy to understand and efficient to operate, there are at least two shortcomings. First, this approach has a low degree of parameter sharing, resulting in a large number of parameters and a huge amount of computation. Second, the relevance of multimodality data is not adequately considered, which will impact the accuracy of the final results.

C. Middle Fusion Methods

Middle fusion methods conduct data fusion at a middle phase when deep features are first extracted from the two modalities and fused subsequently. Therefore, they are also called feature fusion methods.

For example, Chen et al. [29] employed a CNN to extract spectral features from a multispectral image and used another CNN to extract spatial-elevation features from an image transformed from LiDAR data. By fusing the extracted features, logistic regression was used to produce the final classification result. Zhao et al. [30] proposed an effective hierarchical random walk network (HRWN) for joint classification of hyperspectral and LiDAR data. It employed a dual-tunnel CNN to capture spatial features from hyperspectral images, and a pixelwise affinity CNN to capture relationships between classes of different elevations from LiDAR data. The extracted features were then fused and used to predict probability distributions. Mohla et al. [31] proposed a feature fusion and extraction framework (FusAtNet) for collective landcover classification of hyperspectral imagery (HSI) and LiDAR data. It utilized a self-attention mechanism to generate an attention map from HSI modality, and simultaneously a cross-attention approach to harness the LiDAR attention map. The modality-oriented information was joint for carrying out the landcover classification task. Results showed that the proposed FusAtNet achieved the state-of-the-art classification performance. Other feature fusion attempts of optical and LiDAR data included an encoder–decoder translation network by Zhang et al. [9], a reconstruction-based network [11], and a graph fusion network [7].

From the above studies, it can be found that although middle fusion methods conduct data fusion at the feature level, in most cases point features were transformed to image features before performing pixelwise feature fusion. 3-D characteristics of the original LiDAR data, such as unordered, irregular format and permutation invariance, was not fully preserved during the transformation. Besides, it is also challenging to handle the modality discrepancies and balance the contributions of each modality during the feature fusion process.

III. Methodology

Fig. 1 illustrates the overall framework of the proposed method, PIF-Net. It is a two-branch network with multiple information interactions among them. The top part is an image-branch, while the bottom part is a point-branch. Each branch can be divided into two parts, namely, the encoder and the decoder. The proposed PIF-Net composes of four components: multimodal feature extraction, upsampling process, hierarchical feature fusion, and multitask semantic segmentation. The first component corresponds to the encoder of PIF-Net, and extracts deep features from original VHR imagery and ALS point clouds with two branches, separately. The second component corresponds to the decoder of PIF-Net, and upsamples the multimodal features extracted in the first component. In the third component, a hierarchical fusion module with an efficient modality feature interaction mechanism is proposed to adaptively fuse the two high-dimensional features, and outputs fused features in both forms of pixels and points. Finally, per-pixel and per-point multilayer perceptrons (MLPs) are used to obtain pixelwise and pointwise labels, and output semantic segmentation results for VHR imagery and ALS point clouds, separately. Details of the proposed PIF-Net will be illustrated in Sections III-A–III-C.

A. Multimodal Feature Extraction

1) Image Feature Extraction: Image feature extraction is completed by the encoder of the image branch. The input data are VHR imagery, represented by $\mathbf{X} \in \mathbb{R}^{C\times H\times W}$, where C is the number of image channels, and H and W are the height and width of the image, respectively. Internally, the structure of the image feature extraction module is based

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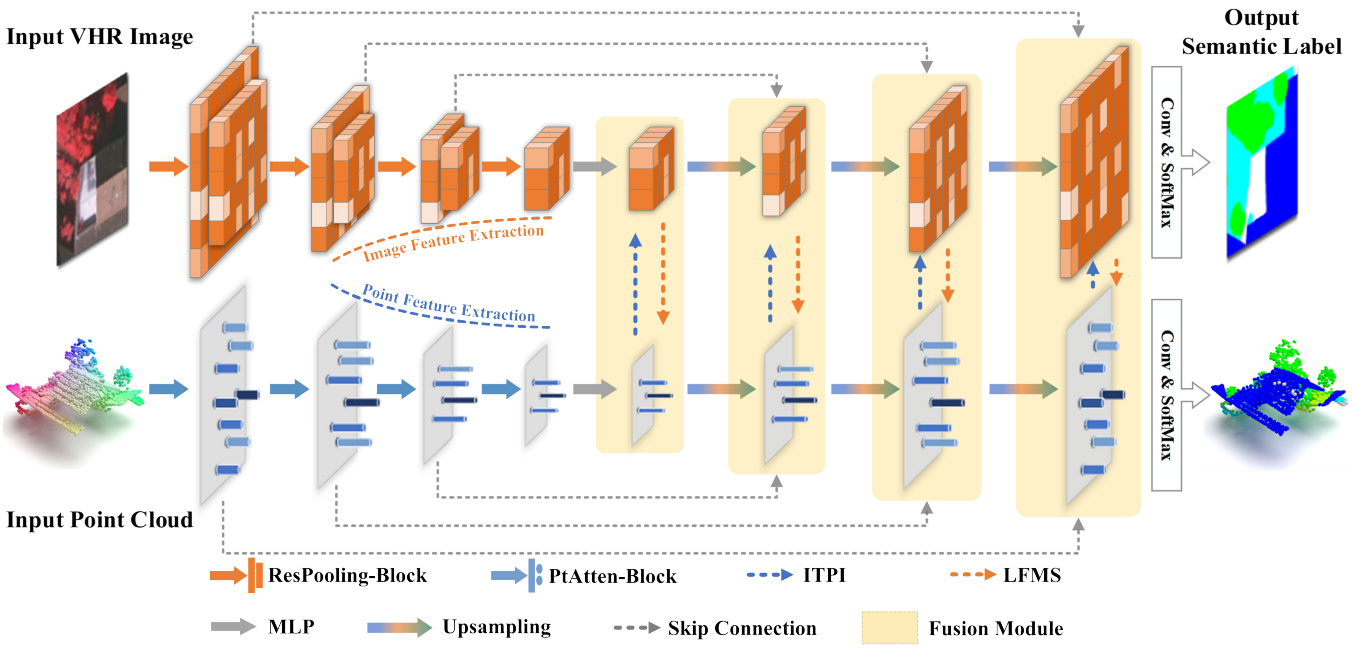


Fig. 1. Architecture of the presented PIF-Net.

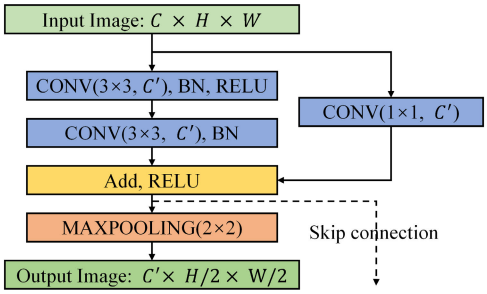


Fig. 2. Architecture of a Res-Pooling block.

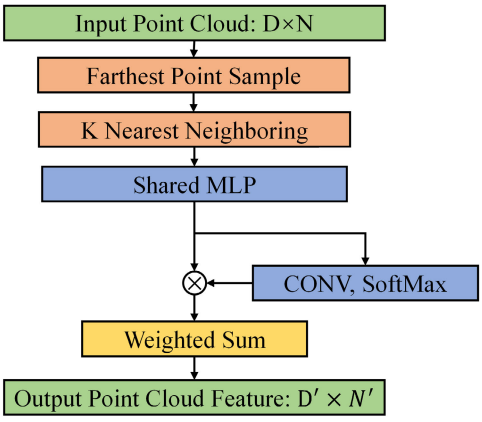


Fig. 3. Architecture of a Pt-Atten block, where ⊗ represents matrix multiplication.

on a U-Net [23] and a ResNet [32] skeleton. It consists of four blocks to extract hierarchical image features, with each block having a residual-like structure followed by a pooling operation. Therefore, we name each block a Res-Pooling block, and its structure is shown in Fig. 2. The front part of Res-Pooling block is two 3 × 3 convolutional layers. We adopt batch normalization (BN) right after each convolution, and a nonlinear activation of rectified linear unit (ReLU) after the first BN. A shortcut connection and pixelwise addition is performed. We adopt the second nonlinear activation after the addition, and it then ends with a max pooling. It is worth mentioning that the last Res-Pooling block does not have the pooling operation (see Fig. 1). Therefore, the output of the ith Res-Pooling block can be represented as

$$\begin{cases} X_i = \text{MaxPooling}\big(R_{\theta_i}(X_{i-1})\big), & i = 1,\ 2,\ 3 \\ X_i = R_{\theta_i}(X_{i-1}), & i = 4 \end{cases} \tag{1}$$

where $R(\cdot)$ represents the Res-Pooling block operation, $X_0 = \mathbf{X}$ represents the original input image, θ_i are the learnable parameters of the ith Res-Pooling block, and MaxPooling$(\cdot)$ the max pooling operation.

After each Res-Pooling block, the height and width of the output image feature map will be reduced by half, and dimension of the image feature map C' is increased. After four Res-Pooling blocks are all completed and performed, an image feature map with small size and high dimensions is obtained.

2) Point Cloud Feature Extraction: The input data for this module are ALS point clouds with 3-D coordinates and intensity information, and can be denoted by $\mathbf{Y} \in \mathbb{R}^{D\times N}$, where N represents the number of points and D represents the dimensions. This module has a similar structure with the encoder of PointNet++ [20], which includes four point attention (Pt-Atten) blocks, shown in Fig. 3. In each Pt-Atten block, we first downsample the input point set using the farthest distance sampling method and keep N_i points, where i is the ith downsampling layer. Second, for each sampled points in the above step, k neighboring points are searched

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with K-nearest-neighboring algorithm and form a point set $Y_i^s \in \mathbb{R}^{D_i \times N_i \times k}$, where D_i represents the number of channels for the sampled points. Third, Y_i^s is fed into an MLP with shared parameters, and it outputs a feature map $y_i \in \mathbb{R}^{d \times N_i \times k}$, where d represents the dimension of each point. Y_i^s and y_i can be represented as

$$Y_i^s = \mathrm{KNN}(Y_{i-1}) \tag{2}$$

$$y_i = \mathrm{MLP}\left(Y_i^s\right) \tag{3}$$

where $\mathrm{KNN}(\cdot)$ represents the K-nearest-neighboring operation, $\mathrm{MLP}(\cdot)$ the MLP operation. We adopt an attention mechanism to enhance the discriminant ability of point feature representations, and each point is assigned with a weight. The weight matrix for the point set Y_i^s is obtained by a convolutional and a softmax operation

$$w_i = \mathrm{SoftMax}(\mathrm{Conv}(y_i)), \quad i = 1, 2, 3, 4 \tag{4}$$

where $w_i \in \mathbb{R}^{N_i \times k}$, and each element $w_i^r (r = 1, 2, \ldots, k)$ represents a weightage for the rth point.

The output of the ith Pt-Atten block is the weighted sum of y_i and w_i, and served as the input of the next Pt-Atten block

$$Y_i = \sum y_i * w_i, \quad i = 1, 2, 3, 4 \tag{5}$$

where $Y_0 = Y$ represents the original input point cloud. After each Pt-Atten block, the number of points is gradually reduced, and at the same time point features are condensed into high-dimensional features. Using four Pt-Atten blocks, a point feature map with sparse points and high dimensions of features is obtained.

B. Upsampling Process

In Section III-A, we obtain an image feature map and a point feature map, with both maps containing high dimensions of features. However, the size of the image feature map becomes smaller and the point feature map becomes sparser than the original inputs. In this section, we adopt upsampling operations to recover the size of the image feature map and the point cloud. The upsampling process corresponds to the decoder of PIF-Net, and can also be divided into the image and point cloud branch. For the image branch, it contains one MLP layer and three upsampling layers, with each upsampling layer implemented by a bilinear interpolation. Assume that the input image feature to an upsampling layer is X^j, the output of the upsampling layer can be represented as

$$X_u^j = \mathrm{BI}\left(X^j\right) \tag{6}$$

where $\mathrm{BI}(\cdot)$ represents the bilinear interpolation operation. After each upsampling layer, the size of the image feature map is doubled, while its dimension is reduced adaptively. When all the upsampling processes are completed, the size of the image feature map is recovered to the original input VHR imagery. For the point cloud branch, it also consists of one MLP layer and three upsampling layers, with each upsampling layer containing an inverse distance weighted (IDW) interpolation operation. Assume the input to an upsampling layer is Y^j, the output of the upsampling layer can be represented as

$$Y_u^j = \mathrm{IDW}\left(Y^j\right) \tag{7}$$

where $\mathrm{IDW}(\cdot)$ represents the IDW interpolation operation. After each upsampling layer, the output point feature map becomes denser, while the dimension of point features decreases. When all the upsampling processes are completed, the size of the point feature map recovers to the original input point cloud.

C. Hierarchical Fusion of Multimodal Features

The aim of this module is to hierarchically fuse the image and point cloud deep features extracted in the above steps. This module consists of three components: point-image mapping, point-pixel feature interacting, and hierarchical fusion module.

1) Point-Image Mapping: To fuse the image and point cloud deep features, we first establish an index mapping between ALS points and VHR image pixels. There are three steps to create such a point-image mapping. First, geographic coordinates of each pixel in VHR imagery are extracted. Pixel coordinates can be either 2-D or 3-D depends on whether height information is available (e.g., height information can be obtained from stereo images). Second, for each point in the point cloud, the nearest pixel in VHR imagery is searched and recorded. To speed the search process, a $k-d$ tree is built for image pixels based on their coordinates. Third, a lookup table is created to store the point-image mapping, with each row storing the index of a point, and a column storing indexes of corresponding pixels.

By creating such a mapping, we ensure that each point has and only has one corresponding pixel. However, each pixel may have zero, one or more than one corresponding points.

2) Point-Pixel Feature Interacting: The outputs of multimodal feature extraction are point- and pixel-based features. Point-based features contain geometric details which are more friendly for identification of instances above the ground (e.g., buildings, trees), while pixel-based features compose of abundant spectral information which effectively distinguish different ground instances. To fully use the information of the multimodality data, feature interaction is required and is the prerequisite for deep feature fusion.

First, point interpolation is conducted to transform point features $\mathbf{Y}$ into an image feature map $\mathbf{X}$. For each pixel x in the image feature map, the number of its corresponding points in $\mathbf{Y}$ is counted using the point-image mapping created in the above step. There are three possible cases: zero, one, and more than one, which correspond to different methods of calculating features of x

$$F_x = \begin{cases} F_{y_i}, & n = 1 \\ \dfrac{1}{n}\displaystyle\sum_{i=1}^{n} F_{y_i}, & n > 1 \\ \mathrm{IDW}\left(F_{\{y_i\}_{x+}^3}\right), & n = 0 \end{cases} \tag{8}$$

where $\{y_i\}_{i=1}^n$ is the set of corresponding points of x, n the number of points in the set, F_x the feature of pixel x, $F_{(y_i)}$ the feature of point y_i, $\{y_i\}_{x+}^3$ the three nearest points in the neighborhood space of x.

After point interpolation is done, a point feature map is transformed to an image feature map, and the resulted feature map can be easily fused with the VHR image feature maps

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by concatenating them. Assume the input point feature map is Y, the output image feature map after point interpolation is represented by ITPI(Y), where ITPI($\cdot$) represents the image transformation operation based on point interpolation (ITPI).

Second, local feature map selection (LFMS) is performed to transform image feature maps to point features. A point set is copied from the point feature map. For each point in the point set, its corresponding pixel in the local image feature map is selected with the created point-image mapping. The pixel feature is then transferred to the specific point. By such a way, a new point feature map with VHR image information is created. After the LFMS operation is completed, a point feature map is obtained, which can be easily concatenated with the ALS point feature map so that the latter contains both VHR image and ALS point clouds information. Assume the input image map is X, the output point map can be represented by LFMS(X), where LFMS($\cdot$) represents the LFMS operation.

3) Hierarchical Fusion Module: The aim of this step is to hierarchically fuse VHR image and ALS point cloud deep features, and output the mixed features in both point and pixel forms. As can be seen in Fig. 1, PIF-Net consists of four fusion modules, namely, FM1, FM2, FM3, and FM4. The structure of FM1 is slightly different from that of others, which are all shown in Fig. 4. Below we explain the four fusion modules, separately.

For the first fusion module (FM1), it includes three steps [see Fig. 4(a)].

First, the image feature map from the last Res-Pooling block (i.e., X_4) is input to a MLP layer, and at the same time, the point feature map from the last Pt-Atten block (i.e., Y_4) is input to another MLP layer. The outputs of MLP layers are represented as

$$X^0,\ Y^0 = \mathrm{MLP}(X_4), \mathrm{MLP}(Y_4). \tag{9}$$

Second, the previous image feature map X^0 is transformed to a point feature map by an LFMS operation, which is followed by a concatenation with the point feature map Y^0. The concatenated point feature map is input to an MLP layer, which outputs a point feature map Y^1

$$Y^1 = \mathrm{MLP}\big[Y^0, \mathrm{LFMS}\big(X^0\big)\big]. \tag{10}$$

Third, the previous point feature map Y^0 is transformed to an image feature map by an ITPI operation, which is followed by a concatenation with the image feature map X^0. The concatenated image feature map is then input to a 1×1 convolutional layer, which outputs an image feature map X^1

$$X^1 = \mathrm{Conv}\big[X^0, \mathrm{ITPI}\big(Y^0\big)\big]. \tag{11}$$

X^1, Y^1 are then input to the next fusion module. For the remaining fusion modules (FM2, FM3, and FM4), each of them includes four steps [see Fig. 4(b)].

Step 1: For the jth fusion module ($j = 2, 3, 4$), in the image branch, it first connects the upsampled image feature X_u^{j-1} (see Section III-B) and the corresponding image feature X_i by the Res-Pooling block (see Section III-A1). The two features should have the same size, and $i + j = 5$. The connected feature is input to a 3×3 convolutional layer, which outputs a mixed image feature

$$\mathrm{MIF}\big(X_i, X_u^{j-1}\big) = \mathrm{Conv}\big[X_i, X_u^{j-1}\big]. \tag{12}$$

Step 2: In the point cloud branch, the jth fusion module connects the upsampled points feature Y_u^{j-1}) (see Section III-B) and the corresponding point feature Y_i from the Pt-Atten block (see Section III-A2). The two features should have the same size, which means the same number of points, and $i + j = 5$. The connected feature is a mixed point feature

$$\mathrm{MPF}\big(Y_i, Y_u^{j-1}\big) = \big[Y_i, Y_u^{j-1}\big]. \tag{13}$$

Step 3: The mixed image feature is transformed to a point feature map by an LMFS operation. The transformed feature is then concatenated with the mixed point feature in Step 2. The concatenated feature is input to an MLP layer, and it outputs a fused feature in point form

$$\mathrm{FUPF}^j = \mathrm{MLP}\big[\mathrm{LFMS}\big(\mathrm{MIF}\big(X_i, X_u^{j-1}\big)\big), \mathrm{MPF}\big(Y_i, Y_u^{j-1}\big)\big]. \tag{14}$$

Step 4: An ITPI operation is performed on the above fused feature in point form to transform it into an image feature map. A residual-like module [see Fig. 4(b)] is then applied to fuse the transformed image feature and the mixed image feature in Step 1, and it outputs a fused feature in pixel form

$$\mathrm{FUIF}^j = F(x, y) + \mathrm{Conv}(x) \tag{15}$$

where $x = \mathrm{MIF}(X_i, X_u^{j-1})$, $y = \mathrm{ITPI}(\mathrm{FUPF}^j)$, $F(x, y) = \mathrm{Conv}[x, y]$.

After the above four steps, the fusion module outputs fused features in both point and pixel forms. The fused feature in pixel form is input to the next upsampling layer in the image branch in Section III-B, i.e., $X^j = \mathrm{FUIF}^j$. The fused feature in point form is input to the next upsampling layer in the point cloud branch in Section III-B, i.e., $Y^j = \mathrm{FUPF}^j$. Steps 1–4 are repeated three times to hierarchically fuse the multimodal

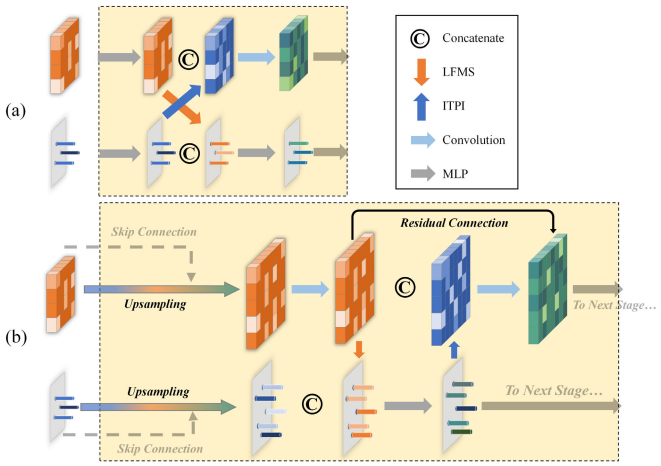


Fig. 4. Architecture of the fusion module. (a) FM1. (b) FM2, FM3, and FM4.

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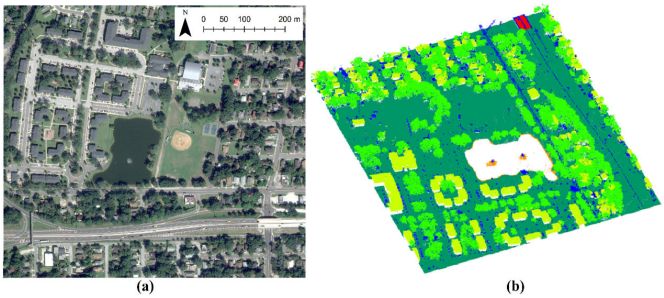


Fig. 5. US3D dataset. (a) VHR imagery. (b) ALS point clouds with ground-truth labels.

features. The last fusion module (i.e., FM4) outputs fused features in both pixel and point forms.

D. Training and Inference for Multitask Semantic Segmentation

After obtaining the multimodal features in point and pixel forms, these features are input to a classification module for semantic segmentation of the ALS point clouds and the VHR imagery. The classification module also has two branches: the image branch and the point cloud branch. The point cloud branch contains a 1-D convolutional layer and a softmax layer. The input to the point cloud branch is the multimodal features in point forms, and it outputs semantic labels of each point. For the image branch of the classification module, it contains a 1×1 convolutional layer and a softmax layer. The input is the multimodal features in pixel forms, while the output is semantic labels of all pixels.

The output of the classification module is subjected to a focal loss function [33] which can better deal with the problem of sample imbalance, and is backpropagated to train the PIF-Net in an end-to-end fashion. The total loss is a weighted sum of two items: the loss of the ALS point clouds and the loss of the VHR imagery

$$\mathrm{LS} = \alpha \sum_{i=1}^{N} -(1-p_i)^{\gamma} \log(p_i) + (1-\alpha) \sum_{j=1}^{M} -\left(1-q_j\right)^{\gamma} \log\left(q_j\right) \quad (16)$$

where M is the number of pixels in the VHR imagery, N the number of points in the ALS point cloud, p_i the predicted probability of the ith point by the proposed PIF-Net, q_j the predicted probability of the jth pixel, and γ the focusing parameter to be tuned, which is recommended to set to 1 or 2. α is the weight for the ALS point cloud, which is between 0 and 1. The value of α will be further discussed in Section V.

IV. Experiments and Results

A. Study Areas and Datasets

Two datasets are used to verify the effectiveness of the proposed network. Both datasets contain VHR imagery and ALS point clouds.

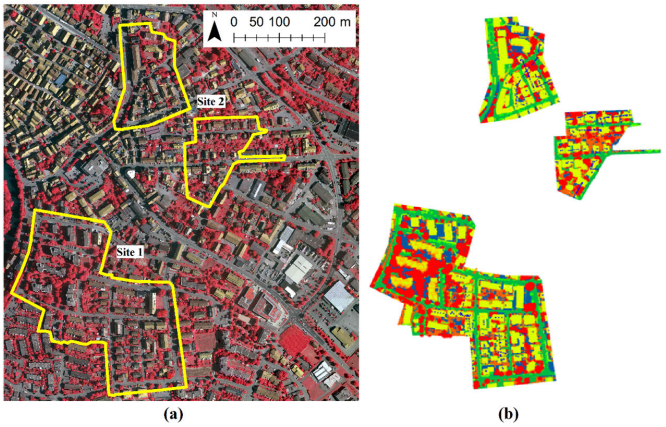


Fig. 6. ISPRS Vaihingen dataset. (a) VHR imagery. (b) ALS point clouds with ground-truth labels.

1) Urban Semantic 3-D Dataset: The Urban Semantic 3-D (US3D) dataset was first released by the 2019 IEEE GRSS Data Fusion Competition (DFC2019) [34]. It covers a total area of approximately 20 km^2 in Jacksonville, Florida, and Omaha, Nebraska, USA. In this study, we only use the dataset in Jacksonville, which is shown in Fig. 5. The US3D includes two data sources: 1) VHR imagery provided by WorldView-3 satellite, which includes a panchromatic image of 0.35-m ground resolution, a four-band multispectral image of 1.4-m ground resolution, an image storing x–y–z coordinates of each pixel in the image, and 2) ALS point clouds acquired by the U.S. Homeland Security Infrastructure Program, which has an average point density of 2 pts/m^2. Ground-truth labels of both VHR imagery and ALS point clouds were provided by organizers based on field surveys, visual inspection, and OpenStreetMap.

Both the VHR imagery and the ALS point clouds are labeled, with each pixel/point falling into one of the five categories: ground, foliage, building, water, and roadway. The study area and the US3D is divided into 10 980 nonoverlapping small portions, with each portion including a 224×224 image, and a point cloud file containing 8192 points after resampling. Correspondences between points and images in each portion are built with a nearest neighborhood method and steps described in Section III-C1. Each pair of an image tile and a point set is served as a whole to train the proposed network.

2) ISPRS Vaihingen Dataset: The ISPRS benchmark is an open access dataset used for 2-D and 3-D semantic labeling contest [35]. This dataset was collected over the city of Vaihingen, Germany. Hereafter, it is referred to as ISPRS Vaihingen dataset (see Fig. 6). The ISPRS Vaihingen dataset comprises of two sites (Site 1 and Site 2), which represents two typical scenes in Vaihingen, Germany. Site 1 covers an area of 399×421 m^2, which is a residential community, with isolated houses and high-rise buildings. Site 2 covers an area of 389×419 m^2, which is in the center of the city, and contains dense and complex buildings.

The ISPRS Vaihingen dataset is divided into 1241 portions, with each portion containing a 224×224 orthoimage with a ground resolution of 0.09 m, and an ALS point cloud of 3072 points with an average point density of 4 points/m^2.

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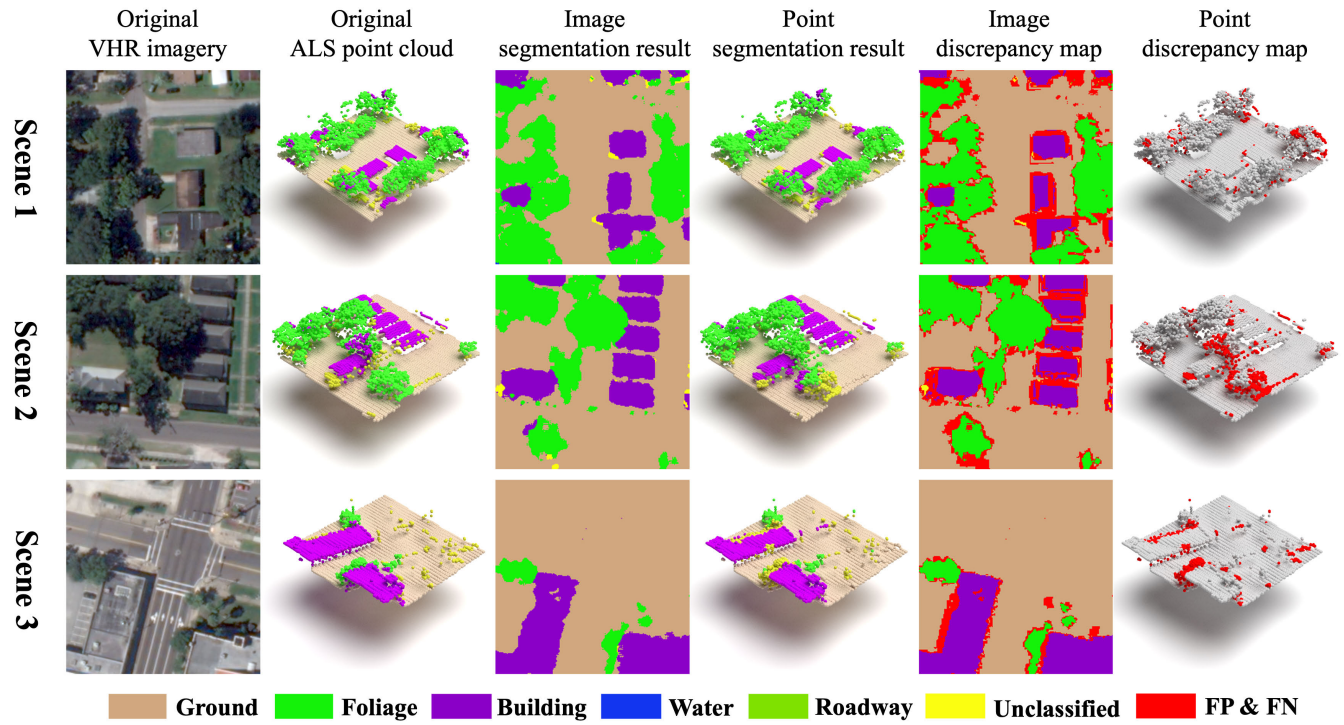


Fig. 7. Detailed segmentation results of three scenes in US3D dataset consisting of VHR imagery with the size of 224 × 224 and ALS point clouds with the size of 8192. (First column) Original VHR imagery. (Second column) Original ALS point cloud. (Third column) Segmentation results of VHR imagery. (Fourth column) Segmentation results of ALS point cloud. (Fifth column) Discrepancies between predicted results and ground truth of VHR imagery. (Sixth column) Discrepancies between predicted results and ground truth of ALS point cloud.

The orthoimages have three bands: Near Infrared, Red, and Green, and were captured by aerial cameras. The ALS point clouds were captured by a Leica ALS50 system, with each point containing x–y–z coordinates, intensity and return count. Both the orthoimage and the ALS point clouds were labeled already. Each point was labeled as one of the following nine classes: powerline, low vegetation, impervious surface, car, fence/hedge, roof, facade, shrub, and tree. Each pixel was labeled as one of the following five classes: low vegetation, impervious surface, car, roof, and tree.

Unlike the IEEE US3D dataset, the VHR images in the Vaihingen dataset are orthoimages, allowing us to directly build correspondences between points and pixels with x–y coordinates and steps in Section III-C1. Similarly, each pair of an orthoimage and a point cloud set is served as a whole to train the network.

B. Experimental Settings

1) Implementation Details: As shown in Fig. 3, both the image and point cloud branches are encoder-decoder architecture, with four downsampling stages, four upsampling stages, and four fusion stages. For image branch, the input image size is 224 × 224, and dimensions of outputs for the above 12 stages are 64, 128, 256, 512, 512, 256, 128, 64, 512, 256, 128, and 64, respectively. For point cloud branch, the input size is 8192 for the US3D dataset and 3072 for the ISPRS Vaihingen dataset, and dimensions of outputs for the above 12 stages are 64, 128, 256, 512, 512, 256, 128, 64, 512, 256, 128, and 64, respectively.

The proposed PIF-Net was trained from scratch in an end-to-end manner. We adopted Adam optimizer, an initial learning rate of 0.001 adjusted by a Cosine Annealing algorithm [36]. The batch size is set as 12, and the epoch is 200. All experiments were performed on a desktop computer with two GPUs of Nvidia RTX 3090, a CPU of AMD Ryzen ThreadRipper 3990X, and 128-GB RAM. The elapsed time for training the US3D dataset is about 10 h, while 7.5 h for training the ISPRS Vaihingen dataset.

2) Evaluation Metrics: We adopt two metrics for accuracy evaluation of each individual class in VHR imagery and ALS point cloud: the F_1 and the intersection over union (IoU). The two metrics are computed as

$$\begin{cases} F_1 = \dfrac{2 \cdot \mathrm{TP}_i}{2 \cdot \mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i} \\ \mathrm{IoU} = \dfrac{\mathrm{TP}_i}{\mathrm{TP}_i + \mathrm{FP}_i + \mathrm{FN}_i} \end{cases} \tag{17}$$

where TP_i is the number of true positive pixels or points for class i, FP_i the number of false positive pixels or points for class i, and FN_i the number of false negative pixels or points for class i. Another two metric are used to assess the accuracy of entire classes in VHR imagery and ALS point cloud: overall accuracy (OA) and mean F_1 (mF_1). The two metrics are computed as

$$\begin{cases} \mathrm{OA} = \dfrac{\sum_{i=1}^{N} \mathrm{TP}_i + \sum_{i=1}^{N} \mathrm{TN}_i}{\sum_{i=1}^{N} \mathrm{TP}_i + \sum_{i=1}^{N} \mathrm{TN}_i + \sum_{i=1}^{N} \mathrm{FN}_i} \\ \mathrm{mF}_1 = \dfrac{1}{N} \sum_{i=1}^{N} F_{1i} \end{cases} \tag{18}$$

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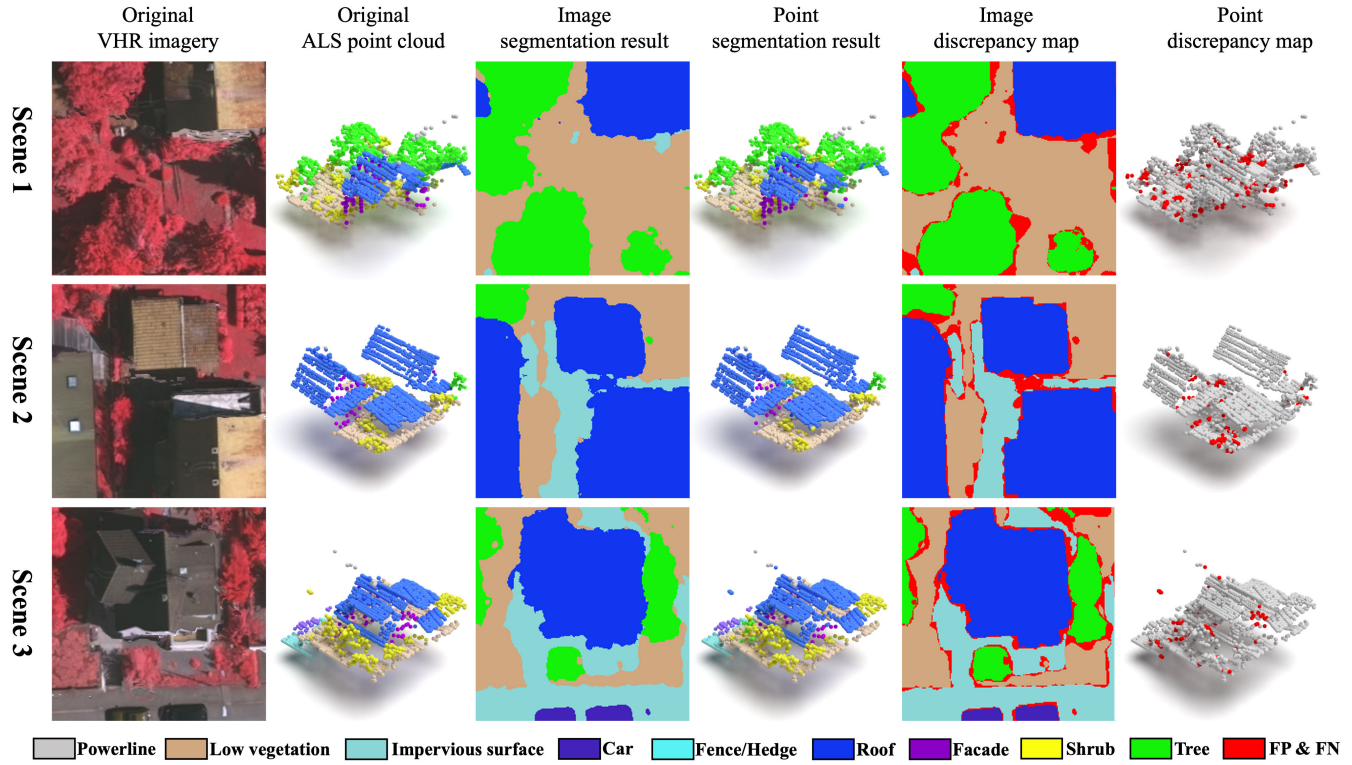


Fig. 8. Detailed segmentation results of three scenes in the ISPRS Vaihingen dataset consisting of VHR imagery and ALS point cloud. (First column) Original VHR imagery. (Second column) ALS point clouds with ground-truth labels. (Third column) Segmentation results of VHR imagery. (Fourth column) Segmentation results of ALS point cloud. (Fifth column) Image discrepancy map with FPs and FNs represented by red. (Sixth column) Point discrepancy map with FPs and FNs represented by red.

where F_{1i} the F_1 value of class i, N the total number of classes.

C. Semantic Segmentation Results

1) Results on US3D Dataset: Part of semantic segmentation results of the US3D dataset are shown in Fig. 7. As can be seen in Fig. 7, the segmentation results by PIF-Net are overall satisfactory, which can be further validated by the quantitative results in Table I. The proposed PIF-Net achieved an OA of 91.5% and an mF_1 of 88.1% in image segmentation, and an OA of 97.2% and an mF_1 of 95.1% in point segmentation. The ALS point clouds have higher segmentation accuracies than the VHR imagery. In terms of individual classes, the IoU and F_1 shows that classes ground and water have higher segmentation accuracies, while classes foliage, building and roadway have relatively lower segmentation accuracies. The class ground has the highest IoU (98.3%) and F_1 (99.1%) on ALS point cloud, and has the second highest IoU (88.8%) and F_1 (94.1%) on VHR imagery. The highest IoU and F_1 on VHR imagery is achieved in class water (91.3% and 95.4%, respectively). The lowest accuracy is achieved in class roadway in both VHR imagery (IoU of 60.7% and F_1 of 75.5%) and ALS point clouds (IoU of 84.4% and F_1 of 91.5%).

The fifth and sixth columns in Fig. 7 show discrepancies between segmentation results by PIF-Net and the ground truth, in which the rendered colors represent FPs and FNs.

TABLE I
QUANTITATIVE SEGMENTATION RESULTS OF US3D DATASET

	VHR imagery (%)		ALS point clouds (%)	
Class	*IoU*	F_1	*IoU*	F_1
Ground	88.8	94.1	98.3	99.1
Foliage	76.8	86.9	91.6	95.6
Building	78.5	87.9	85.7	92.3
Water	91.4	95.4	93.8	96.8
Roadway	60.7	75.5	84.4	91.5
OA = 91.5 mF_1 = 88.1			OA = 97.2 mF_1 = 95.1	

According to Fig. 7, the main misclassification occurred along boundaries of roofs and foliage.

2) Results on ISPRS Vaihingen Dataset: In ISPRS Vaihingen dataset, Site 1 was employed as the training set to train the proposed PIF-Net model, after which, the model was tested on Site 2. We have demonstrated semantic segmentation results of part of the ISPRS Vaihingen dataset in Fig. 8, in which the last two columns show discrepancies between predicted results and the ground truth on VHR imagery and ALS point cloud. As can be seen in Fig. 8, most of pixels and points were correctly classified. Quantitatively, the proposed PIF-Net achieved an OA of 90.1% and an mF_1 of 88.0% for VHR imagery, and an OA of 89.3% and an mF_1 of 79.2% for ALS point clouds (see Table II). Compared to the US3D dataset, most of the metrics on VHR imagery and ALS point clouds were lower. There are several possible reasons to explain this phenomenon. First, the training and testing datasets in the

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TABLE II
QUANTITATIVE SEGMENTATION RESULTS OF ISPRS VAIHINGEN DATASET

VHR imagery (%)			ALS point clouds (%)	
Class	IoU	F_1	IoU	F_1
Low vegetation	76.3	85.7	83.4	91.1
Impervious surface	84.6	90.3	86.9	92.7
Powerline	-	-	58.0	72.4
Car	69.2	83.7	63.1	77.7
Fence/Hege	-	-	47.0	63.2
Roof	89.3	93.5	92.6	96.1
Facade	-	-	46.4	64.4
Shrub	-	-	55.0	67.2
Tree	79.4	87.0	80.7	88.3
OA = 90.1			OA = 89.3	
mF_1 = 88.0			mF_1 = 79.2	

US3D dataset were from the same area, while in the ISPRS Vaihingen dataset, they were from different sites, leading to larger heterogeneity between training and testing samples. Second, the available training samples in the ISPRS Vaihingen dataset were much less than those in the US3D dataset, which is mainly because most of points in the ISPRS Vaihingen dataset were without ground-truth labels, and common areas with ground-truth labels of both pixel and point were even much less. Third, in the ISPRS Vaihingen dataset, some classes were available on ALS point cloud, while were missing on VHR imagery (e.g., powerline, fence/hedge, facade, shrub), leading to poorer performance of these classes.

For the remaining classes, the highest IoU was achieved for class roof on both VHR imagery (89.3%) and ALS point clouds (92.6%). The lowest IoU was achieved for class car on both VHR imagery (69.2%) and ALS point clouds (63.1%). The poor performance for class car can be attributed to the different acquisition time of VHR imagery and ALS point cloud, and the characteristic of moving, causing cars in the two data sources were not completely matchable. According to the image discrepancy map in Fig. 8, most of the wrongly labeled pixels occurred along boundaries of objects. One potential reason for such a phenomenon maybe attributed to voids or gaps when generating ortho-images. In addition, significant confusions were observed between class low vegetation and tree. These two classes have similar spectral characteristics, which makes it difficult to distinguish on VHR imagery. Although height information in the ALS point clouds is useful to distinguish the above two classes, it is difficult to determine the class of vegetation with medium height.

V. DISCUSSION

A. Parameter Sensitivity Analysis of the Weight for Multimodal Data

In Section III-D, a weight of α for the ALS point clouds was adopted to calculate the loss function, while the weight for the VHR imagery is $1-\alpha$. A larger weight indicates that the specific single-modal data play a more significant role in the multitask semantic segmentation experiment. To evaluate how the parameter of α impacts the performance of the proposed PIF-Net, a number of experiments were conducted on the ISPRS Vaihingen dataset with different α values of 0, 0.2, 0.4, 0.5, 0.6, 0.8 and 1. An α value of 0 means that only the VHR imagery was used in the network, while an α value of 1 means that only the ALS point clouds were used in the network. In the above two cases, the presented PIF-Net becomes a single-modality network.

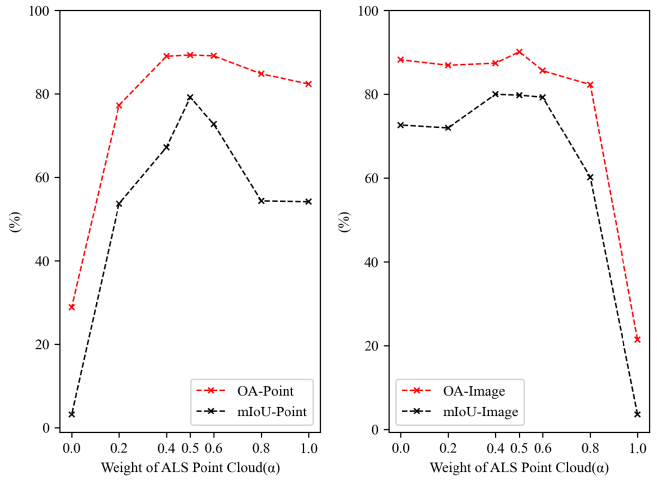


Fig. 9. Impacts of weight of ALS point clouds (α) on segmentation results. (Left) Impacts of α on image segmentation. (Right) Impacts of α on point segmentation.

Fig. 9 illustrates the OAs of semantic segmentation for both ALS point cloud and VHR imagery with different α values. It is observed that the OA for VHR imagery decreases all the way as α increases, and the OA for ALS point clouds first increases as α increases ($\alpha < 0.5$), and reaches the maximum (89.41%) when α is 0.5, and decreases as α continues to increase. It is worth mentioning that the fluctuation of OA for VHR imagery is not significant when α is between 0.2 and 0.6. Therefore, we consider that the optimal α value for the ISPRS Vaihingen dataset should be set to 0.5 so that satisfactory segmentation results of both image and point cloud can be obtained. However, users can specify the α value according to the actual demand, and a large α value should be used when the quality of ALS point clouds is much better than that of VHR imagery. Similarly, a large weight of VHR imagery should be used in the following cases: 1) segmentation results of VHR imagery are much more important than those of ALS point cloud and 2) the quality of VHR imagery is much better than that of ALS point cloud. Specifically, we analyzed the two cases of $\alpha = 0$ and $\alpha = 1$, in which the multimodality network becomes a single-modality network.

The OA is 21.4% for VHR imagery and 82.4% for ALS point clouds when $\alpha = 1$ and the presented network becomes a single-modality network of point cloud. Compared to $\alpha = 0.5$ (OA = 90.1% for VHR imagery and OA = 89.3% for ALS point cloud), the OA for ALS point clouds is 6.9% lower. The results indicated that the presented PIF-Net significantly improved the semantic segmentation results obtained by the single-modality network of point cloud, which proved the effectiveness of the presented network.

In the second case of $\alpha = 0$, the presented network becomes a single-modality network of imagery, and the OA is 88.2% for VHR imagery and 28.9% for ALS point cloud. The OA for ALS point clouds is very low as the height

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TABLE III

QUANTITATIVE COMPARISONS OF THE PROPOSED PIF-NET WITH POPULAR SINGLE-MODALITY AND MULTIMODALITY APPROACHES FOR IMAGE AND POINT CLOUD SEGMENTATIONS ON THE ISPRS VAIHINGEN DATASET

Type	Modality	Method	Extra Data	Image Class					Point Cloud Class									OA	mF_1
				Roof	Tree	Low veg	Car	Imp sur	Powerline	Low veg	Imp sur	Car	Fence/ Hedge	Roof	Facade	Shrub	Tree		
Image Based	Single-Modality	UNetFormer	-	**95.3**	**90.6**	**84.9**	**88.5**	**92.7**	-	-	-	-	-	-	-	-	-	**91.0**	**90.3**
		FuseNet	-	92.1	84.9	84.3	85.4	90.8	-	-	-	-	-	-	-	-	-	88.7	87.5
		MAResU-Net	-	94.8	90.0	79.1	85.9	92.2	-	-	-	-	-	-	-	-	-	90.2	88.5
		OCRNet	-	93.8	87.1	81.8	82.2	90.5	-	-	-	-	-	-	-	-	-	88.4	87.1
		PSPNet	-	93.8	86.7	80.9	81.7	90.2	-	-	-	-	-	-	-	-	-	88.0	86.7
		DeepLab V3+	-	92.2	86.5	80.8	77.6	89.5	-	-	-	-	-	-	-	-	-	87.7	85.3
		UperNet	-	92.0	85.6	80.2	78.3	88.8	-	-	-	-	-	-	-	-	-	87.2	85.0
		U-Net	-	91.3	83.9	77.8	64.9	87.7	-	-	-	-	-	-	-	-	-	84.5	81.1
	Multi-Modality	DSA2F	DSM	95.1	90.7	86.4	88.3	**92.5**	-	-	-	-	-	-	-	-	-	**92.8**	**90.6**
		MFNet	DSM	96.3	89.1	84.7	**89.7**	92.2	-	-	-	-	-	-	-	-	-	91.7	90.4
		CMFNet	DSM	95.8	89.9	86.2	87.5	91.7	-	-	-	-	-	-	-	-	-	91.6	90.1
		SAGATE	DSM	94.8	92.6	81.3	87.8	91.7	-	-	-	-	-	-	-	-	-	91.1	89.8
		TransUNet	DSM	**96.5**	**92.8**	76.1	69.6	91.7	-	-	-	-	-	-	-	-	-	91.0	87.3
		ESANet	DSM	95.7	90.5	77.2	85.5	91.4	-	-	-	-	-	-	-	-	-	90.6	88.2
		FuseNet	DSM	96.3	90.3	79.0	81.4	91.7	-	-	-	-	-	-	-	-	-	90.5	87.7
		MKE	DSM	92.6	89.6	**88.2**	83.5	92.7	-	-	-	-	-	-	-	-	-	90.0	89.3
		PSPNet	DSM	94.5	90.2	78.8	79.2	92.0	-	-	-	-	-	-	-	-	-	89.9	86.6
		ABMDRNet	DSM	93.7	88.3	82.9	87.1	91.8	-	-	-	-	-	-	-	-	-	89.2	88.8
		DeepLab V3+	DSM	93.1	84.6	81.9	78.1	90.1	-	-	-	-	-	-	-	-	-	88.6	85.6
		U-Net	DSM	92.3	83.8	81.2	63.8	87.3	-	-	-	-	-	-	-	-	-	85.4	81.7
Point Cloud Based	Single-Modality	PointNet	-	-	-	-	-	-	52.6	70.0	83.2	11.2	7.5	74.8	7.8	24.6	45.4	65.7	41.9
		PointNet++	-	-	-	-	-	-	57.9	79.6	90.6	66.1	31.5	91.6	54.3	41.6	77.0	81.2	65.6
		PointSift	-	-	-	-	-	-	55.7	80.7	90.9	77.8	30.5	92.5	56.9	44.4	79.6	82.2	67.7
		D-FCN	-	-	-	-	-	-	70.4	80.2	91.4	78.1	37.0	93.0	60.5	46.0	79.4	82.2	70.7
		PointCNN	-	-	-	-	-	-	61.5	82.7	91.8	75.8	35.9	92.7	57.8	49.1	78.1	83.3	69.5
		KPConv	-	-	-	-	-	-	63.1	82.3	91.4	72.5	25.2	94.4	60.3	44.9	81.2	83.7	68.4
		GADH-Net	-	-	-	-	-	-	66.8	82.5	91.5	78.3	35.0	94.6	63.3	49.8	**83.9**	**85.0**	71.7
		alsNet	-	-	-	-	-	-	70.1	80.5	90.2	45.7	7.6	93.1	47.3	34.7	74.5	80.6	60.4
		WhuY4	-	-	-	-	-	-	42.5	**82.7**	91.4	74.7	**53.7**	94.3	53.1	47.9	82.8	84.9	69.2
		RandLA-Net	-	-	-	-	-	-	69.0	80.1	91.4	73.1	36.2	93.7	58.4	45.8	82.7	83.1	70.1
		GraNet	-	-	-	-	-	-	67.7	82.7	91.7	**80.9**	51.1	**94.5**	62.0	49.9	82.0	84.5	73.6
		LGENet	-	-	-	-	-	-	**76.5**	82.1	**91.8**	80.0	40.6	93.8	**64.7**	**49.9**	83.6	84.5	**73.7**
	Multi-Modality	PointNet++	RGB	-	-	-	-	-	61.4	84.9	**91.3**	70.4	36.6	90.4	57.4	49.9	76.9	84.1	68.4
		RandLA-Net	RGB	-	-	-	-	-	**71.5**	**85.2**	91.1	**70.9**	**59.4**	**92.8**	**60.1**	**51.3**	**83.9**	**86.0**	**74.0**
Point-Image Fusion	Multi-Modality	PIF-Net	-	93.5	87.0	85.7	83.7	90.3	72.4	91.1	92.7	77.7	63.2	96.1	64.4	67.2	88.3	90.1 / 89.3	88.0 / 79.2

information is missing without ALS point cloud. Compared to $\alpha = 0.5$, the OA for VHR imagery is 1.9% lower. According to the above results, the presented PIF-Net slightly increase the OA of image semantic segmentation obtained by the single-modality network. Additionally, it was observed that image segmentation results of class tree were increased by 1.57% after setting the weight of ALS point clouds to 0.5. We can expect better improvements of image segmentation by introducing more training samples in PIF-Net.

B. Comparisons With Existing Methods

To evaluate the effectiveness of the proposed PIF-Net, we further compared our results with existing popular single-modality methods including image segmentation approaches and point cloud segmentation approaches. To ensure the fairness of the comparisons, we adopted the same evaluation metric F_1 for each individual class, and OA and mF_1 for all classes.

Comparisons were made in two aspects. First, we compared PIF-Net with single-modality methods, where only one modality data were involved. Either Image segmentation or point cloud segmentation was conducted on the single-modality data without auxiliary data sources. Second, comparisons were made between PIF-Net and multimodality methods, where both images and point clouds were involved in the segmentation process. Quantitative comparison results were shown in

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TABLE IV
QUANTITATIVE COMPARISONS OF THE PROPOSED PIF-NET WITH POPULAR SINGLE-MODALITY AND MULTIMODALITY APPROACHES FOR IMAGE AND POINT CLOUD SEGMENTATIONS ON THE IEEE US3D DATASET

Type	Modality	Method	Image Class						Point Cloud Class					OA	mF_1
			Extra Data	Ground	High Veg	Building	Water	Bridge Deck	Ground	High Veg	Building	Water	Bridge Deck		
Image Based	Single-Modality	DeepLabV3+	-	73.2	77.3	84.2	95.9	94.2	-	-	-	-	-	84.4	85.0
		U-Net	-	68.2	82.2	91.3	97.5	93.8	-	-	-	-	-	81.0	84.6
		FCN	-	73.8	78.5	84.7	95.3	94.2	-	-	-	-	-	84.7	85.3
		PSPNet	-	59.2	75.8	**92.6**	**98.0**	**97.5**	-	-	-	-	-	85.4	84.6
		LANet	-	76.9	80.0	87.6	95.9	95.3	-	-	-	-	-	86.8	87.1
		SwinUNet	-	69.2	76.6	57.0	94.3	44.1	-	-	-	-	-	68.7	68.3
		TransUNet	-	**81.4**	**83.9**	91.7	97.5	94.8	-	-	-	-	-	**89.6**	**89.8**
	Multi-Modality	DeepLabV3+	DSM	**87.2**	**85.3**	88.3	94.6	72.2	-	-	-	-	-	84.4	85.5
		U-Net	DSM	83.6	85.6	**92.2**	**96.6**	**75.3**	-	-	-	-	-	**86.9**	**86.7**
Point cloud Based	Single-Modality	PointSIFT	-	-	-	-	-	-	98.6	97.0	85.5	46.4	60.4	94.0	77.6
		PointCNN	-	-	-	-	-	-	98.7	97.2	84.9	44.1	65.3	93.8	78.0
		KPConv	-	-	-	-	-	-	98.4	94.2	87.4	43.0	77.5	94.5	80.1
		D-FCN	-	-	-	-	-	-	99.1	98.1	89.9	45.0	73.0	95.6	81.0
		LGGAA	-	-	-	-	-	-	98.9	96.1	90.2	41.6	83.7	94.8	81.4
		GACNN	-	-	-	-	-	-	99.3	96.8	91.1	42.5	84.4	95.1	82.8
		PointNet++	-	-	-	-	-	-	98.3	95.8	79.7	44.4	47.3	92.7	73.1
		DA-Net	-	-	-	-	-	-	99.3	97.6	92.7	41.6	**85.1**	98.3	83.3
		LGENet	-	-	-	-	-	-	**99.3**	**98.3**	92.8	47.4	79.1	**98.4**	83.4
		DGCNN	-	-	-	-	-	-	97.9	93.2	90.4	**88.2**	54.4	95.1	84.2
		DANCE-NET	-	-	-	-	-	-	99.1	93.9	87.0	58.3	83.9	96.8	84.4
		PointConv	-	-	-	-	-	-	97.3	95.8	**93.6**	74.5	69.2	95.3	86.1
		RandLA-Net	-	-	-	-	-	-	97.1	93.6	92.6	78.4	76.2	96.8	**87.6**
	Multi-Modality	PointNet++	RGB	-	-	-	-	-	**97.6**	94.7	82.2	44.6	**54.8**	93.0	74.8
		RandLA-Net	RGB	-	-	-	-	-	97.2	**94.8**	**89.9**	**57.4**	84.7	**96.4**	**84.8**
Point-Image Fusion	Multi-Modality	PIF-Net	-	94.1	86.9	87.9	95.4	75.5	99.1	95.6	92.3	96.8	91.5	91.5 / 97.2	88.1 / 95.1

Tables III and IV, in which metrics for the whole class (OA and mF_1) and individual classes (F_1) have been recorded.

1) Comparisons With Single-Modality Methods: For the ISPRS dataset, we first compared PIF-Net with image segmentation approaches including UNetFormer [37], FuseNet [38], MAResU-Net [39], OCRNet [40], PSPNet [41], DeepLabV3+ [24], UperNet [42], and U-Net [23]. PIF-Net obtained an OA of 90.1% and an mF_1 of 88.0%, which were slightly lower than the highest. However, PIF-Net produced the best results in some individual classes. For example, it achieved the highest F_1 in class low_veg (85.7%), which was 0.8% than the second highest.

Second, we compared PIF-Net with classic point cloud segmentation approaches including PointNet [19], PointNet++ [20], PointSIFT [43], D-FCN [44], PointCNN [45], KPConv [46], GADH-Net [47], alsNet [48], WhuY4 [49], RandLA-Net [50], GraNet [51], LGENet [52]. Other than PIF-Net, the highest OA and mF_1 were 85.0% and 73.7%, and were obtained by GADH-Net and LGENet, respectively. Meanwhile, PIF-Net achieved an OA of 89.3% and an mF_1 of 79.2%, significantly higher than existing methods.

For the US3D dataset, PIF-Net was also compared with single-modalities methods in terms of both images and point clouds. Image segmentation approaches conducted on the US3D dataset include DeepLabV3+ [24], U-Net [23], FCN [53], PSPNet [41], LANet [54], SwinUNet [55], TransUNet [56], while point cloud segmentation approaches include PointSIFT [43], PointCNN [45], KPConv [46], D-FCN [44], LGGAA [57], GACNN [58], PointNet++ [20], DA-Net [59], LGENet [52], DGCNN [60], DANCE-NET [61], PointConv [62], and RandLA-Net [50]. According to Table IV, it shows that PIF-Net achieved the highest OA (91.5%) for image segmentation and the highest mF_1 (95.1%) for point cloud segmentation. The extraordinary performance proved the superiority of our proposed PIF-Net over existing single-modality methods.

2) Comparisons With Multimodality Methods: To further demonstrate the effectiveness of our proposed data fusion strategy, we compared PIF-Net with multimodality methods. Early fusion strategy was employed in these methods, which incorporated DSMs in image segmentation or incorporated RGB information in point cloud segmentation (see Tables III and IV). For the ISPRS dataset, PIF-Net was first compared with 12 existing methods incorporating DSMs in image segmentation. These methods include DSA2F [63], MFNet [64], CMFNet [65], SAGATE [66], TransUNet [56], ESANet [67],

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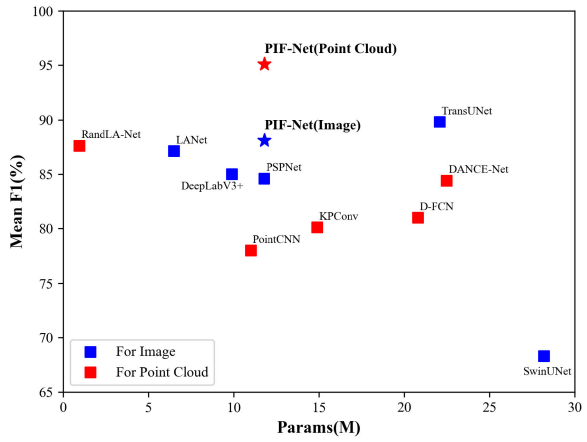


Fig. 10. Number of parameters of different point and image classification methods. Blue and red represent image and point classification methods, respectively.

FuseNet [38], MKE [68], PSPNet [41], ABMDRNet [69], DeepLab V3+ [24], and U-Net [23]. Among all the 13 methods, the OA of PIF-Net (90.1%) ranks eight, which seems to be an ordinary performance in image segmentation. In terms of point cloud segmentation, two additional experiments of incorporating RGB information were introduced using methods of PointNet++ and RandLA-Net. It can be seen in Table III that the OA and mF_1 by PointNet++ was significantly improved after adding RGB information into point clouds. However, they were still 3.3% and 5.2% lower than PIF-Net. Moreover, F_1 scores for all individual classes by PIF-Net were higher than those of other multimodality methods. The above results demonstrated that our proposed PIF-Net performed significantly better than existing multimodality methods in terms of point cloud segmentation, while achieved the moderate level in terms of image segmentation.

Similar results were found on the US3D dataset. OA of PIF-Net was slightly better than that of U-Net and DeepLab V3+ in image segmentation, while metrics of PIF-Net were significantly higher than those of other multimodality methods in point cloud segmentation.

C. Network Complexity

We compare existing point cloud and image classification networks with the proposed PIF-Net in two dimensions: the number of parameters and the classification accuracy. As shown in Fig. 10, PIF-Net achieves a good balance between classification accuracy and network complexity in both image and point cloud segmentation. Its classification accuracy is among the best while the number of parameters is at a moderate level.

VI. Conclusion

In this article, we have proposed a deep point-image fusion network, namely PIF-Net, for multimodality semantic segmentation, where joint deep feature representations are learned from both the input modalities of ALS point clouds and VHR imagery. The proposed PIF-Net comprises four components. The first component extracts multimodal deep features using two independent branches, in which Res-Pooling blocks are proposed and used in the image branch, and Pt-Atten blocks are proposed and used in the point branch. It outputs condensed features with high dimensions. The second component upsamples the extracted multimodality features. In the third component, a hierarchical fusion module with an efficient modality feature interaction mechanism of ITPI and LFMS is proposed to adaptively fuse features of different levels and varying modalities. It outputs joint features in both the point and pixel representations, which are further input to MLPs in the fourth component to obtain multimodality semantic segmentation results.

Two experiments were performed on the IEEE US3D dataset and the ISPRS Vaihingen dataset. Both qualitative and quantitative results by the proposed approach were overall satisfactory. Parameter sensitivity analysis was conducted to assess contributions of each modality data and obtain suitable weights. Comparisons with existing single-modality and multimodality methods have indicated that PIF-Net exceeded most classic methods and could bring significant improvements. In terms of network complexity, the proposed PIF-Net achieved better performance with a good balance between less number of parameters and higher accuracies.

Although the quantitative evaluations demonstrated effectiveness of the proposed PIF-Net, there are at least two limitations. First, the proposed PIF-Net is built on the assumption that the multimodality data are precisely coregistered. However, this is not always true, and the proposed method may suffer from coregistration problems. Second, the proposed method requires users to manually input weights for ALS point clouds and VHR imagery, which will impact the multimodality segmentation results greatly. In the future, we will develop a registration-free approach which can automatically choose the optimal weights and reduce human interventions to the greatest extent.

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